

EduPro Predictive Analytics — EDA

1. Verification Status

The original EDA was substantially consistent with the underlying files. The corrected version below keeps the verified findings, removes potentially misleading wording, and makes the forecast split and model-performance interpretation explicit. All numerical values were recomputed from the uploaded source workbooks where possible.

Check	Verified result
Users	3,000
Teachers	60
Courses	60
Transactions	10,000
Missing values	0 across all four source tables
Duplicate rows	0 across all four source tables
Invalid UserID/CourseID/TeacherID references	0 / 0 / 0
Historical transactions	7,551
Future/test transactions	2,449

2. Dataset and Forecast Split

The platform contains 3,000 users, 60 teachers, 60 courses and 10,000 transaction events. The temporal split is reproducibly obtained at 2025-10-01: 7,551 transactions occur before the split and 2,449 occur on/after it. The historical period generated \$684,084.58, while the future observation period generated \$227,238.89. The course-level forecast file contains 60 observations with historical features and future targets.

Important wording correction: the 2,449 records are transaction/enrollment events, not 2,449 unique students. The analysis therefore reports them as future enrollments/transactions unless unique-user counts are explicitly calculated.

3. Course Structure and Pricing

Measure	Verified result
Total courses	60
Free courses	38 (63.33%)
Paid courses	22 (36.67%)
Subject categories	12
Courses per category	5 each
Paid price range	\$0.78 – \$490.90
Mean paid price	\$253.60

4. Instructor Workload Concentration

TC00042 and TC00040 account for 6,086 of 10,000 transaction events (60.86%). The source workbook confirms 3,061 transactions for TC00042 and 3,025 for TC00040. This is a genuine concentration signal, although transaction concentration should not be interpreted as evidence of teaching quality or individual instructor performance.

5. Forecast Results

Measure	Actual future	Predicted	Aggregate accuracy
Future enrollment	2,449	2,464.385	99.3718%
Future revenue	\$227,238.89	\$222,744.35	98.0221%

The aggregate accuracy figures are platform-level comparisons between summed predictions and summed future observations.

They must not be presented as the same thing as course-level R^2 or as proof that every course prediction is highly accurate.

6. Model Evaluation Interpretation

Target	Selected model	MAE	RMSE	R^2
Future enrollment	Random Forest	6.130417	7.322472	-0.236758
Future revenue	Gradient Boosting	\$814.003	\$1,457.665	0.960063

For enrollment, Random Forest is selected because it has the lowest RMSE and highest R^2 among the evaluated enrollment models, even though its R^2 is negative. Gradient Boosting has slightly lower MAE but higher RMSE and lower R^2 . The negative enrollment R^2 indicates weak course-level out-of-sample explanatory power and should remain visible in the EDA.

For revenue, Gradient Boosting is the selected model with $R^2 = 0.960063$. The research paper explains that this strong result is closely related to the separation between zero-revenue free courses and paid courses and to historical financial features.

7. Freemium Dynamics

Metric	Free (38)	Paid (22)	Total
Actual future enrollments	1,571	878	2,449
Enrollment share	64.15%	35.85%	100.00%
Actual future revenue	\$0.00	\$227,238.89	\$227,238.89
Predicted future enrollment	1,564.755	899.630	2,464.385
Predicted future revenue	\$0.08	\$222,744.27	\$222,744.35

Verified finding: free courses generate 64.15% of future enrollment events but no direct revenue, while paid courses generate 100% of direct future revenue. This supports treating enrollment and monetization as separate analytical targets.

8. Category-Level Future Revenue

Category	Actual future revenue
Artificial Intelligence	\$52,438.14
Business	\$46,278.51
Project Management	\$44,356.45
Programming	\$17,946.64
Data Science	\$17,577.97
Cybersecurity	\$16,164.66
Digital Marketing	\$10,910.35
Design	\$10,778.43
Finance	\$6,536.24
Web Development	\$4,228.10
Machine Learning	\$23.40
Marketing	\$0.00

9. High-Opportunity Courses — Clarified

The original EDA, research paper, and dashboard consistently identify CR00040 (Leadership in Projects), CR00006 (Graphic Design Basics), and CR00024 (Big Data Analytics) as the project's three-course high-opportunity set. That statement is retained for consistency. However, these are NOT the top three courses by predicted revenue: CR00012 (Business Strategy), CR00001 (Python Basics), and CR00040 (Leadership in Projects) rank highest by predicted revenue. Therefore, the corrected EDA describes the three-course set as a project-defined portfolio recommendation based on combined demand and financial yield, rather than calling it the 'top three by predicted revenue.'

Course	Type	Pred. enrollment	Pred. revenue
CR00040 — Leadership in Projects	Paid	46.235	\$15,743.50
CR00006 — Graphic Design Basics	Paid	43.985	\$12,247.90
CR00024 — Big Data Analytics	Paid	44.015	\$5,295.59

10. Feature Importance and Pricing Interpretation

The research reports that Past Revenue (49.97%), Course Price (25.35%), and Past Revenue Per Enrollment (20.44%) account for 95.76% of the Gradient Boosting revenue model's feature importance. This should be interpreted as predictive association, not causation. The reported approximately 0.9902 correlation between course price and predicted revenue is strongly influenced by the free-versus-paid structure and does not establish that raising prices will increase revenue.

11. Final Verified Findings

- The underlying source data contains 3,000 users, 60 teachers, 60 courses and 10,000 transactions with zero missing values, zero duplicates and valid foreign-key relationships.
- The historical/future transaction split is 7,551 / 2,449 and reproduces the revenue totals used in the research.
- Free courses account for 64.15% of future enrollment events and \$0 direct future revenue; paid courses account for 100% of direct future revenue.
- TC00042 and TC00040 together account for 60.86% of all transactions.
- Random Forest is the selected enrollment model, but its course-level R^2 is negative (-0.236758); the 99.37% figure is aggregate forecast agreement, not course-level R^2 .
- Gradient Boosting is the selected revenue model with $R^2 = 0.960063$ and aggregate forecast agreement of 98.02%.
- The three-course high-opportunity set is retained but explicitly labeled as a project-defined combined demand/financial-yield recommendation, not a top-three revenue ranking.

Source basis: EduPro Online Platform workbook; EduPro Forecast workbook; Final Research Paper EduPro Predictive Modeling; EduPro Streamlit Dashboard Evidence. No external data were introduced.